2023 - quiz3

Graded

Student

William Chen

Total Points

10 / 10 pts

Question 1

A large... **5** / 5 pts

1.1 **In units...** **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** calculation error
- **– 2 pts** problem not solved

1.2 **Now someone...** **3** / 3 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** calculation error
- **– 2 pts** approach error
- **– 3 pts** problem not solved

Question 2

A company... **5** / 5 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** calculation error
- **– 3 pts** wrong approach
- **– 4 pts** problem not solved
- **– 2 pts** approach error

Quiz 03

PH-UY 2023, Spring 2026
A Sections

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(1) A large open storage tank (no lid) collects rain water at the top of a hill. A long pipe runs downhill from the tank and eventually runs horizontally along the ground. The water level in the tank is 20 m above the ground. Initially, all the water in the tank and the pipe is stationary.

(a) (2 pts) In units of atmospheres, what is the ***gauge pressure*** at ground level in the pipe? You can use $g \approx 10\,\text{m/s}^2$ for ease if needed, but you should be able to answer this with zero calculation!

$P = P_0 + \rho g h$

At ground level, $\rho g h = P - P_0$ → gauge pressure.

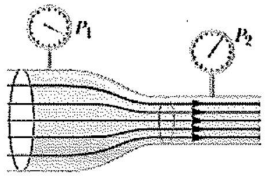
height is 20 m, so in units of atmosphere, gauge pressure → 2 ATM

(b) (3 pts) Now someone punctures the pipe at ground level, making a tiny hole (with area much smaller than the open area of the tank). Water spurts out, and eventually stabilizes into a steady upward stream. Ignore air resistance on the stream. **Approximately how high above the ground is the top of the stream? Be sure to justify your answer.** *Again, if you're calculating a lot, you're failing to see the big picture.*

$P = F/A$

At the top of the stream

The height reaches to the point at the top of the tank because the pressure applied from the 20 m = $\rho g h$ as the gauge pressure.



(2) (5 pts) A company uses a Venturi tube to measure the flow rate of an ideal liquid with a specific gravity of 3/5. The inlet tube (tube 1) has double the diameter of the outlet tube (tube 2), and the difference in the pressures at the centers of the inlet and outlet tubes is $P_1 - P_2 = 0.05$ atm.

Find the speed of the liquid at the center of the outlet tube (tube 2). To simplify the math, take 1 atm $\approx 10^5$ Pa and the density of water to be $10^3\,\text{kg/m}^3$.

$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$

$P_1 - P_2 = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2 \quad P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$

$5000 P_a = \frac{1}{2}(600)(15 v_1^2)$

$v_1^2 = 1.11\,m^2/s^2$

$v_1 = 1.054\,m/s$

$v_2 = 4 \cdot 1.054\,m/s = 4.216\,m/s$

$A_1 v_1 = A_2 v_2$

$\frac{A_1}{A_2} = \frac{v_2}{v_1}$

$4 = \frac{v_2}{v_1}$

$v_2 = 4 v_1$